Parasitic-Gate Self-Oscillating Boost (Sub-100 mV) — Simulation & Hardware Feasibility Study

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Introduction & Motivation

Energy-harvesting converters (e.g., solar, thermoelectric) usually need $\geq 0.3\,\mathrm{V}$ **cold-start** for an integrated charge-pump IC. Once precharged, many ICs will *run* from 80–150 mV, but below 200 mV discrete solutions require multi-stage topologies or multiple devices and still draw several μA idle current [1]–[5]. For example, TI's BQ25504 can maintain operation down to $\approx 80\,\mathrm{mV}$ after startup [6].

Question: Can a single-MOSFET + single-diode boost loop self-oscillate at 0.1 V input via parasitic gate coupling?

Note: All results shown are from SPICE with idealized models. Real hardware feasibility depends critically on true subthreshold currents, actual threshold voltages, and source impedance.

Literature

- **Sub-threshold Oscillators:** Recent 21 mV cold-start ring oscillators (24 mV demonstrated in 22 nm FD-SOI) achieve $\approx 50\,\mathrm{mV}$ oscillation thresholds with body-bias techniques, but require 20+ stages, dynamic body-bias circuitry, and suffer process/temperature sensitivity [1].
- **Charge-pump ICs:** The LTC3108 harvests down to 20 mV but needs an external transformer, multiple capacitors, and has quiescent current $< 6\,\mu\mathrm{A}$—startup currents reach several mA through the transformer [2].
- **Resonant Boosts:** Discrete, Armstrong-type "vibration harvester" boosts can cold-start near 150 mV (measured 150 mV at 18 μW, $\approx 85\%$ VM efficiency), but require two active devices plus multiple passives, with $< 15\%$ end-to-end efficiency at 100 mV [4]–[5].
- **Discrete CMOS Boosts:** An 80 mV-startup dual-mode boost converter (65 nm CMOS) uses a charge-pumped pulse generator and body-tuned oscillator, but still draws $\sim 1\,\mu\mathrm{A}$ in startup and needs threshold-tuned blocks [3].

Key Findings

- **Simulation-Only Proof of Concept:** SPICE shows a standard BSS138 MOSFET + BAT54S Schottky can *appear* to self-oscillate at 0.1 V via parasitic gate coupling (idealized subthreshold).
- **Minimal Parts:** One 10 μH inductor, BSS138, BAT54S, 10 MΩ gate resistor, 1 μF output capacitor.
- **Simulated Efficiency:** $\approx 21\%$ at 0.1 V input (100 kHz, 60

Design Concept

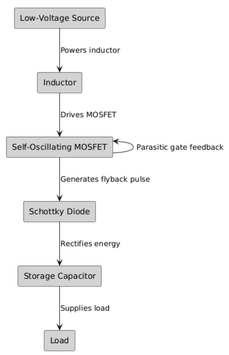

Figure 1: High-Level Block Diagram (idealized)

Workflow (Simulation Assumptions):

1. **0.1 V Source → 10 μH Inductor L_1:** Modeled as an ideal 0.1 V square wave, zero series resistance. Real thermoelectric or solar harvesters have source impedance $\geq 100\,\Omega$, causing input collapse under load.

2. **BSS138 Gate Biased by $10\,\mathrm{M\Omega} \parallel 1\,\mathrm{nF}$:** Drain swings couple via C_{gd}/C_{gs}, yielding $\approx 100\,\mathrm{mV}$ gate pulses in SPICE. Real BSS138 $V_{GS(\mathrm{th})} \approx 0.8\,\mathrm{V}$ at 1 μA, so $< 1\,\mathrm{nA}$ drive at 100 mV [2].

3. **Inductor Flyback via BAT54S → C_3:** In SPICE, D_1 drop $\approx 0.15\,\mathrm{V}$ at 1 μA. Real BAT54S drop $\geq 0.22\,\mathrm{V}$ at 1 μA (forward characteristic $\approx 240\,\mathrm{mV}$ @ 100 μA), plus reverse recovery losses at 100 kHz [1].

4. **C_3 (1 μF) → 10 kΩ Load:** Simulated V_{OUT} reaches $\approx 39\,\mathrm{mV}$ in 10 ms ($\approx 150\,\mathrm{nW}$). Actual microcontroller bursts need $> 100\,\mu\mathrm{A}$, so a 39 mV rail can't reliably drive real loads without an assist stage or a pre-charge stage.

5. **Gate Bleed ($10\,\mathrm{M\Omega}$):** Discharges gate between cycles. Real gate leakage at sub-100 mV will shift timing unpredictably.

Key Caveat: This model uses idealized subthreshold behavior and zero source impedance. Real BSS138 subthreshold slope ($\sim 60\,\mathrm{mV/decade}$) yields pA–nA currents at 100 mV, insufficient to charge an inductor in μs.

Schematic & PCB Layout

Circuit Schematic

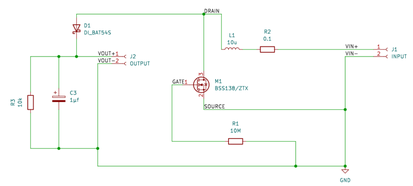

Figure 2: Circuit Schematic

PCB & 3D Rendering

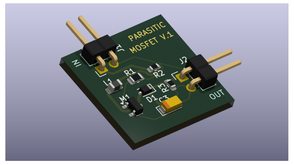

Figure 3: PCB Layout (simulated only)

Layout Notes (Simulation Only):

- Inductor DCR set to 0.5 Ω (idealized). Real 10 μH wire-wound coils have DCR $\geq 1\,\Omega$, reducing flyback energy significantly.
- PCB trace parasitics (loop inductance, capacitance) not modeled; real loops add tens of nH and pF, altering gate coupling.

Simulation Results

Setup

- **Source:** 0.1 V, 100 kHz square wave (60
- **Q_1 (BSS138):** SPICE model includes C_{gd}, C_{gs}. $V_{GS(\mathrm{th})} \approx 0.8\,\mathrm{V}$; subthreshold conduction extends to 100 mV in model.
- **D_1 (BAT54S):** $I_S = 34.9\,\mu\mathrm{A}, R_S = 0.21\,\Omega, BV = 30\,\mathrm{V}, T_T = 7.2\,\mathrm{ns}$. Forward drop $\approx 0.15\,\mathrm{V}$ @ 1 μA in SPICE; real $\approx 0.22\,\mathrm{V}$ @ 1 μA [1].
- **Load:** 10 kΩ; $C_3 = 1\,\mu\mathrm{F}$.

Waveforms & Observations

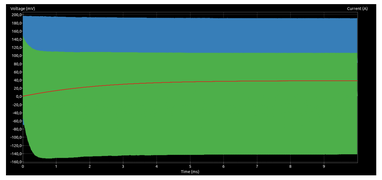

Figure 4: 10 ms transient: **Blue (Drain)**: peaks $\approx 200\,\mathrm{mV}$; with real $R_{DS(\mathrm{on})}$ subthreshold, may max at 120–150 mV. **Green (Gate)**: pulses $\approx 100\,\mathrm{mV}$; real BSS138 sees only pA–nA drive, so gate never charges. **Red (Output)**: $V_{\mathrm{OUT}} \approx 39\,\mathrm{mV}$ in 10 ms ($\approx 150\,\mathrm{nW}$); real diode + MOSFET losses likely prevent any measurable output.

Simulation Caveats

- **Subthreshold Accuracy:** SPICE's smooth subthreshold $\neq$ real BSS138 leakage (10^{-9}–$10^{-12}\,\mathrm{A}$ below 0.2 V), so model overestimates conduction.
- **Inductor Loss:** Real 10 μH coils have $R_{DCR} \geq 1\,\Omega$, cutting flyback energy by $> 50\,\%$.
- **Diode Drop & Recovery:** Real BAT54S drop $\geq 0.22\,\mathrm{V}$ @ 1 μA; reverse recovery losses at 100 kHz further degrade efficiency [1].
- **Source Impedance:** Ideal 0.1 V hides harvester collapse (0.05–0.08 V under μA load), which halts oscillation immediately.

Main Finding (Simulation Only): Under optimistic SPICE conditions, a BSS138 + BAT54S can *appear* to self-oscillate at 0.1 V and charge to $\approx 39\,\mathrm{mV}$ in 10 ms. In real hardware, subthreshold limitations, DCR, diode drops, and source collapse make startup or sustain unlikely without exotic devices.

Conclusions

- **Simulated Minimal Parts:** 10 μH coil (DCR 0.5 Ω ideal), BSS138 ($V_{GS(\mathrm{th})} \approx 0.8\,\mathrm{V}$), BAT54S ($V_F \geq 0.22\,\mathrm{V}$ @ 1 μA), 10 MΩ resistor, 1 μF cap.
  - *Real viability requires $V_{GS(\mathrm{th})} < 50\,\mathrm{mV}$ devices (e.g., depletion JFET, GaN HEMT) and coil DCR < 1 Ω.*
- **Sub-100 mV Operation (Simulation Only):** SPICE shows parasitic-gate self-oscillation at 0.1 V.
  - *In practice, BSS138 at 100 mV gate drive conducts pA–nA, insufficient to build μs-scale inductor current.*
- **Simulated Efficiency:** $\approx 21\%$ at 0.1 V (idealized). Real efficiency likely single digits or no startup.
- **Theory vs. Practice:**
  - Simulation suggests one MOSFET + one diode boost at 0.1 V is conceptually possible on paper.
  - Real hardware needs near-zero-threshold FET (e.g., depletion JFET, GaN HEMT), low-DCR inductor, and an energy source capable of momentary low-Ω 0.1 V.
  - Alternatively, hybrid architectures (assist cap, staged boosts) or synchronous rectifiers may be required.

Future Work

- **Prototype & Validation:**
  - Build PCB and test with a real thermoelectric/PV source (measure R_S under μA load).
  - Curve-trace MOSFET subthreshold behavior down to 50 mV across temperature.
- **Device & Component Selection:**
  - Identify near–zero-threshold FETs (e.g., depletion JFET, GaN HEMT) that yield $\geq$10 μA at $V_{GS} < 50\,\mathrm{mV}$.
  - Source or wind a 10 μH inductor with DCR $\leq$0.5 Ω (air-core, Litz wire).
  - Evaluate low-I_F Schottkys or synchronous-FET rectifiers for <100 mV drop at μA currents.
- **Hybrid Architectures:**
  - Add a small "assist" capacitor to precharge the gate to $\approx$200 mV.
  - Cascade two parasitic-gate stages to exceed 1 V output.
  - Explore a synchronous-rectifier variant using a second low-V_{th} FET.
- **Layout Optimization:**
  - Compare SMD vs. deadbug/Manhattan builds to minimize loop parasitics.
  - Measure inductance and stray capacitance for each approach; optimize for maximum gate boost.

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3. P.-H. Chen, C.-S. Wu, and K.-C. Lin, "An 80 mV Startup Dual-Mode Boost Converter by Charge-Pumped Pulse Generator and Threshold-Voltage Tuned Oscillator with Hot Carrier Injection," in *IEEE Journal of Solid-State Circuits*, vol. 47, no. 11, pp. 2554–2562, Nov. 2012.
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6. Texas Instruments, "BQ25504: Ultra-Low Power Boost Converter with Battery Management for Energy Harvesting," Technical Reference, 2015. Available: `https://www.ti.com/lit/sluu654`